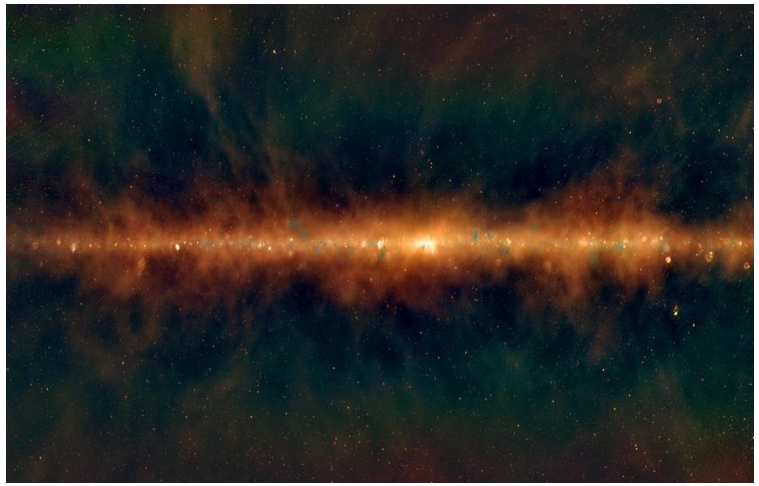
Radio Astronomy: Studying the universe at the lowest frequencies

Bijaya Luitel

Research Assistant, Curtin Singapore/ICRAR Curtin

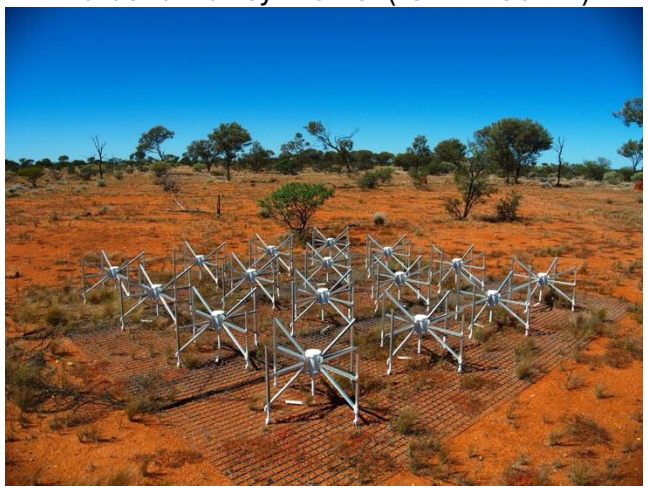
Natasha Hurley-Walker (ICRAR/Curtin) and GLEAM



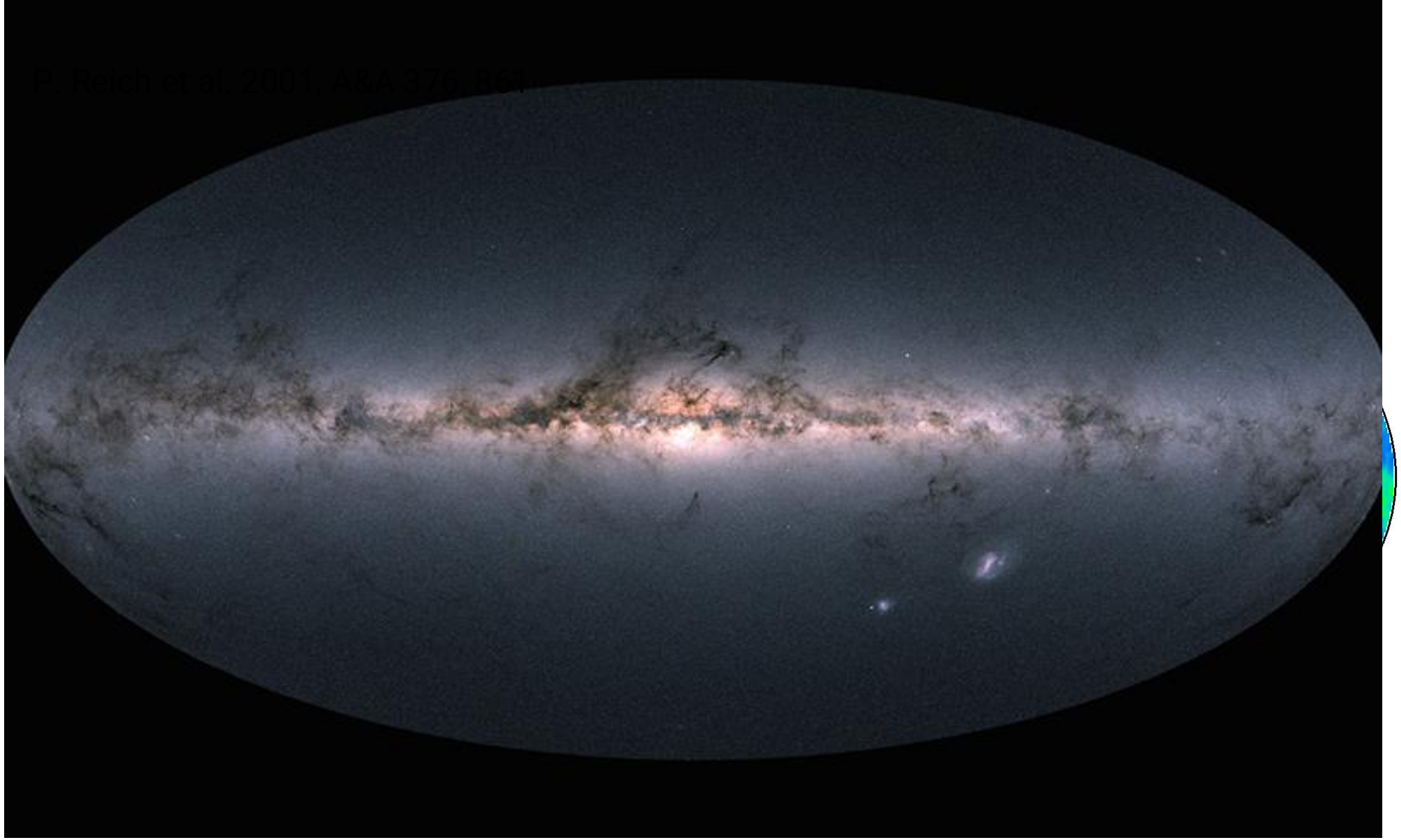
ALMA



Natasha Hurley-Walker (ICRAR/Curtin)



P. Reich et al. 2001, A&A 376, 861



Stockert 25-m and Villa Elisa 30-m

Image Credit: AFP/ESA

Brief Recap: Light

- Light is an electromagnetic wave.
- What we think of light—visible light—is just a small fraction of the EM spectrum.
- Astrophysicists today use the entirety of the EM spectrum to study the universe.

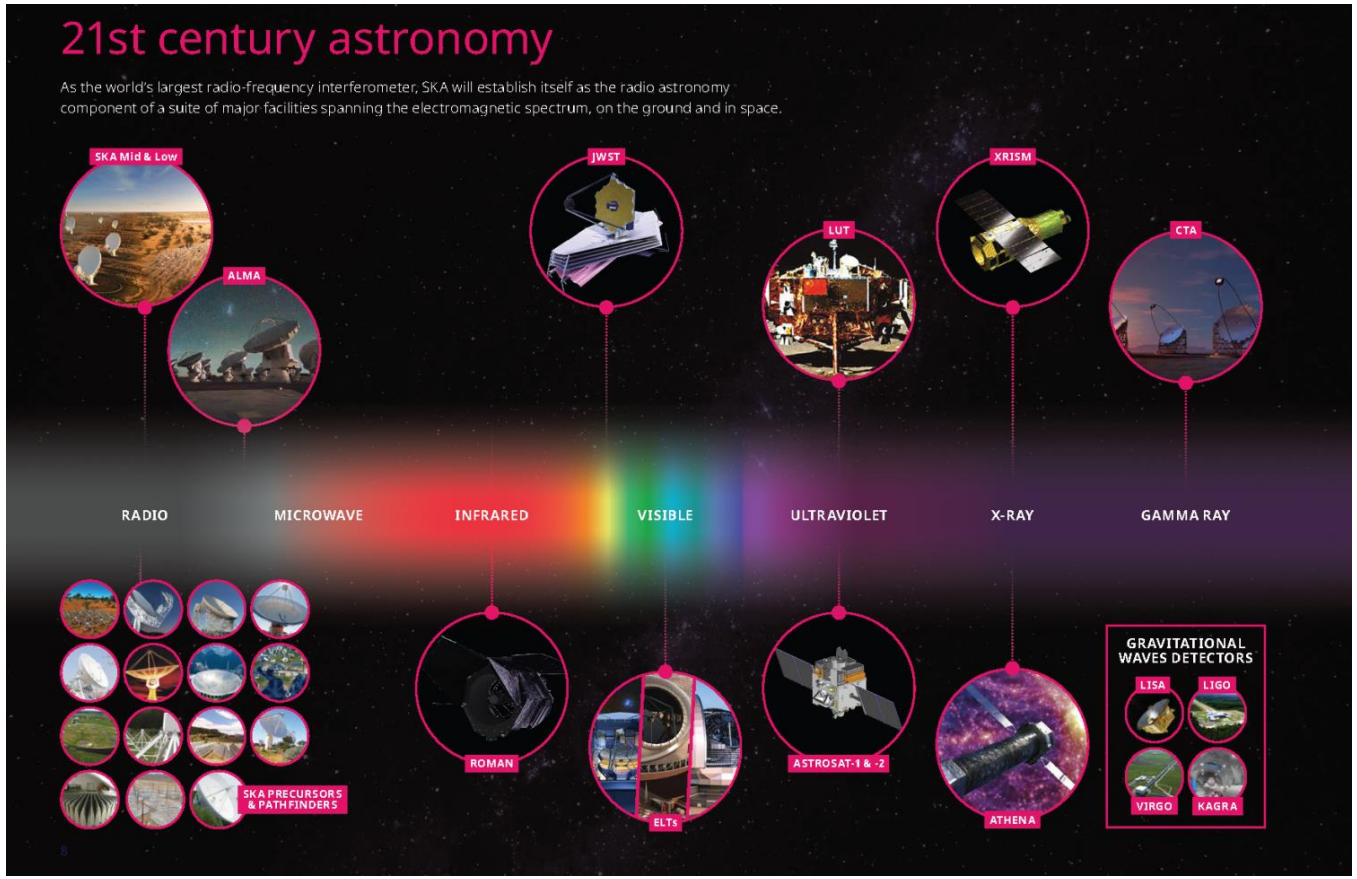
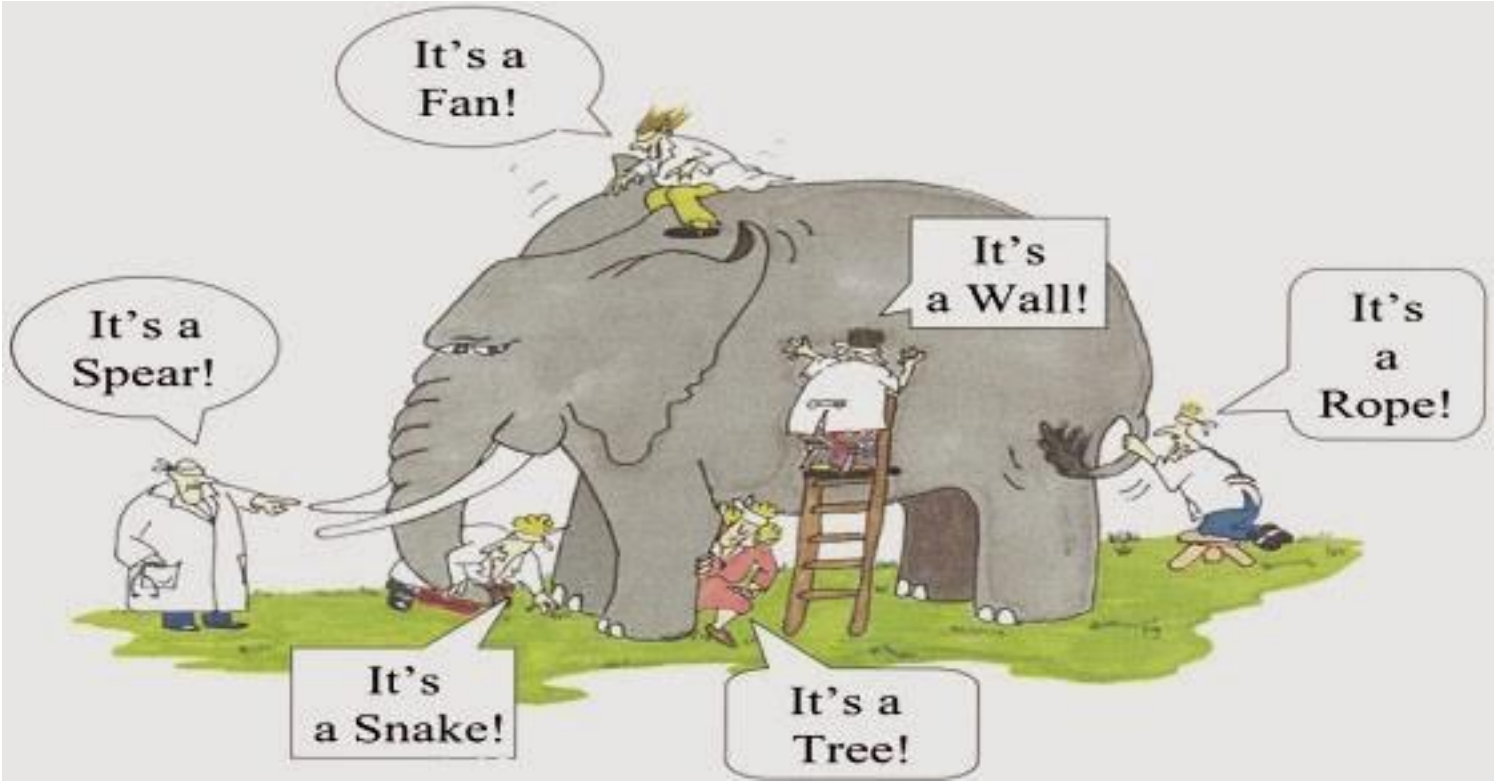


Image Credit: SKAO

Why use different wavelengths?

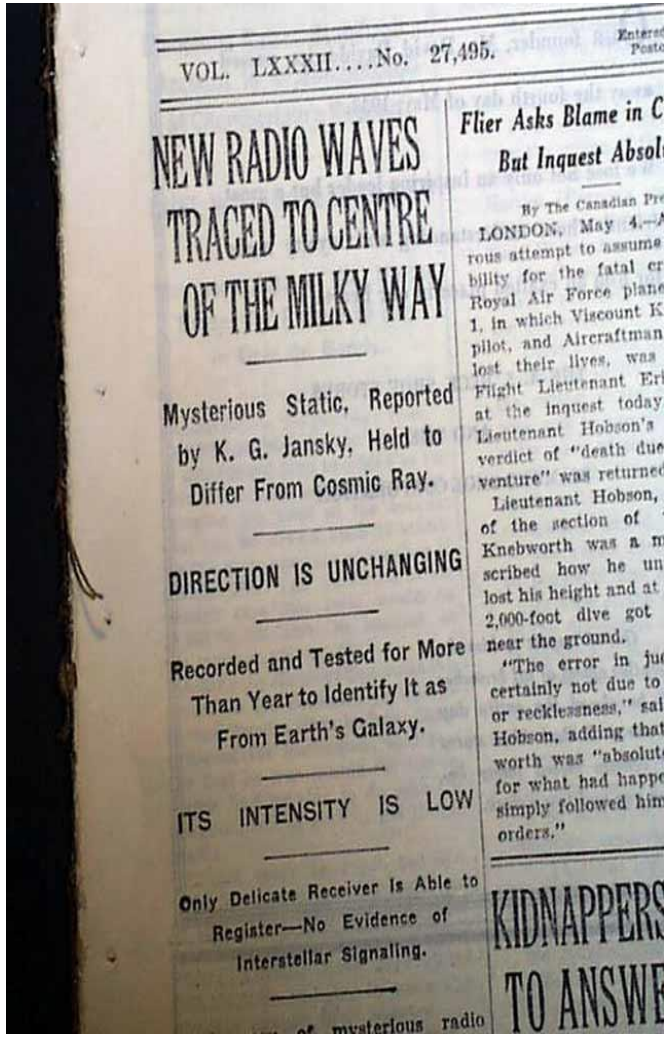


To get a complete picture of a physical object, you need to look at all the possible information it may convey!

Beginnings

- Radio “Astronomy” began with the unintentional discovery of Sag A*
- By Karl Jansky at Bell Labs, in 1933.
- In the next 90 years, we have gone from that to imaging that source of radio waves: the SMBH at the center of our galaxy.

Credit: NYTimes



VOL. LXXXII....No. 27,495.

NEW RADIO WAVES TRACED TO CENTRE OF THE MILKY WAY

Mysterious Static, Reported by K. G. Jansky, Held to Differ From Cosmic Ray.

DIRECTION IS UNCHANGING

Recorded and Tested for More Than Year to Identify It as From Earth's Galaxy.

ITS INTENSITY IS LOW

Only Delicate Receiver Is Able to Register—No Evidence of Interstellar Signaling.

Flier Asks Blame in C
But Inquest Absol

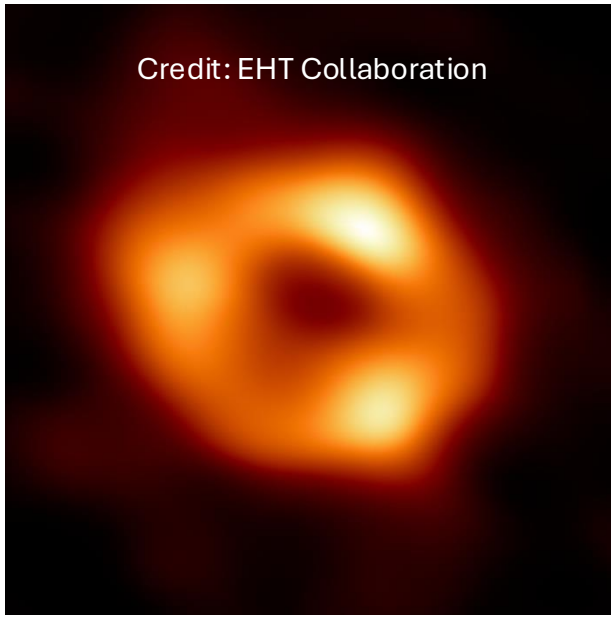
By The Canadian Pre

LONDON, May 4.—A
rous attempt to assume
bility for the fatal cr
Royal Air Force plane
1, in which Viscount K
pilot, and Aircraftman
lost their lives, was
Flight Lieutenant Eri
at the inquest today
Lieutenant Hobson's
verdict of "death due
venture" was returned
Lieutenant Hobson,
of the section of
Knebworth was a m
scribed how he un
lost his height and at
2,000-foot dive got
near the ground.
"The error in ju
certainly not due to
or recklessness," sai
Hobson, adding that
worth was "absolut
for what had happe
simply followed him
orders."

KIDNAPPERS
TO ANSWE

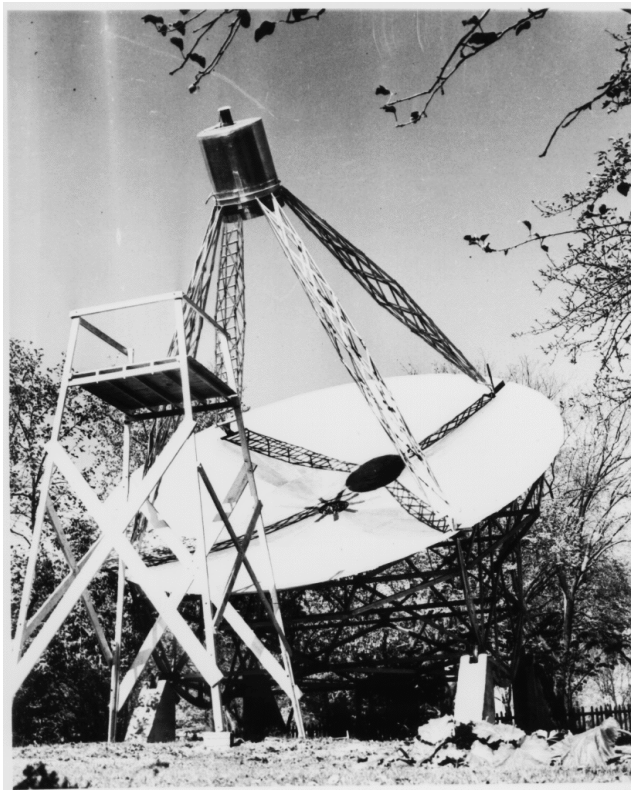
Credit: NRAO/AUI/NSF



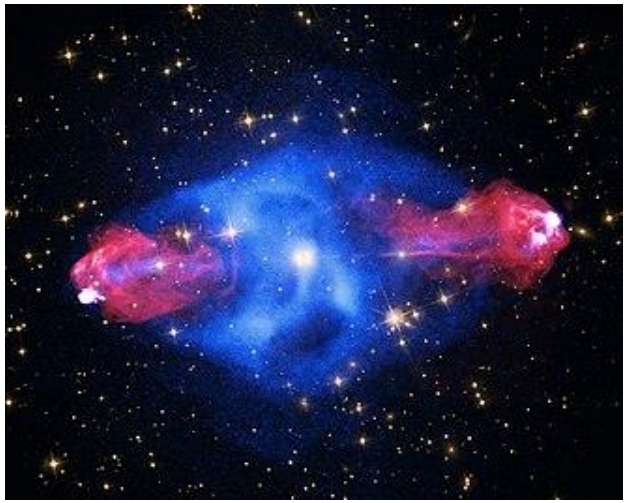


Reber's Radio Telescope

- Grote Reber designed and built the first astronomical radio telescope.
- He made it in his backyard in 1937.
- Using it, he discovered some of the most powerful radio sources in the sky, including Cygnus A.



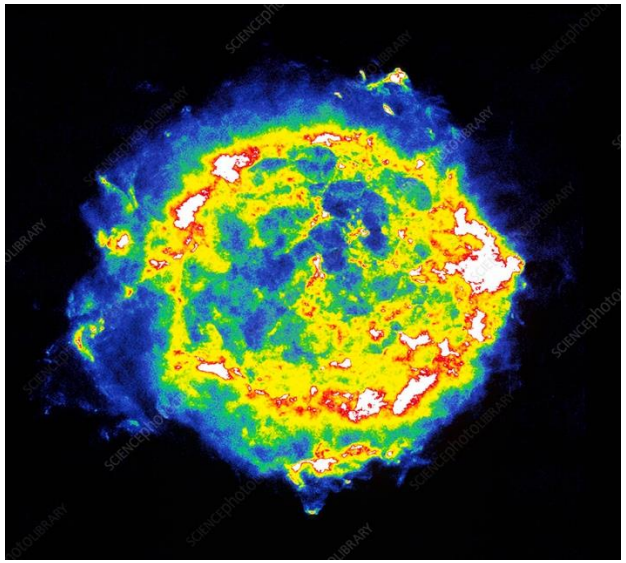
Grote Reber's telescope



Cygnus A: Blue X-rays, Red Radio Waves, and Yellow Optical
Credit: CXC/SAO/NASA/NRAO

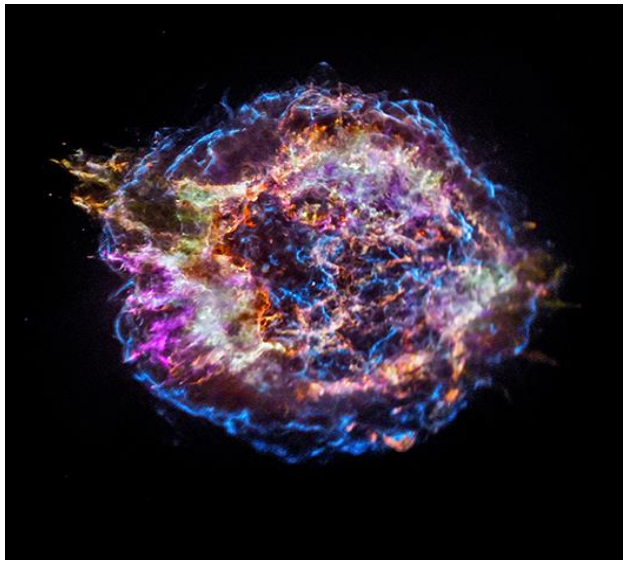
How are radio waves produced?

- Rapidly moving electrons, when they get accelerated (like in powerful magnetic fields)
- By spectral line transitions
- As thermal radiation by low temperature dust clouds in space
- Artificial sources like our satellite communications, radar



Cas A in Radio; red most intense, blue least intense

Credits: M Anderson et al./NRAO



Cas A in X-rays; silicon (red), sulfur (yellow), calcium (green) and iron (purple)

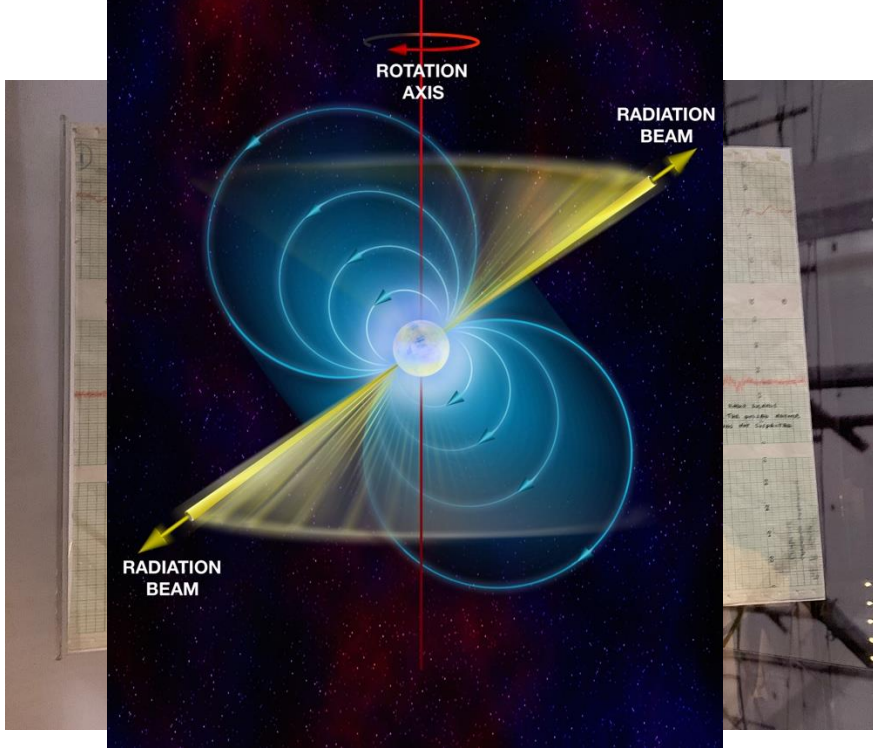
Credits: NASA/CXC/SAO

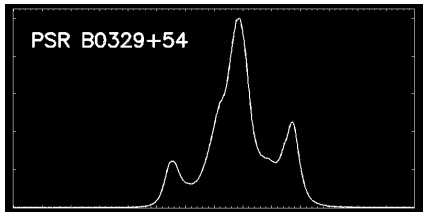
Cool scientific discoveries

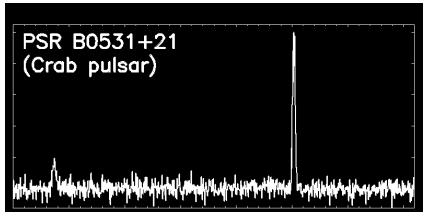
Curtin University

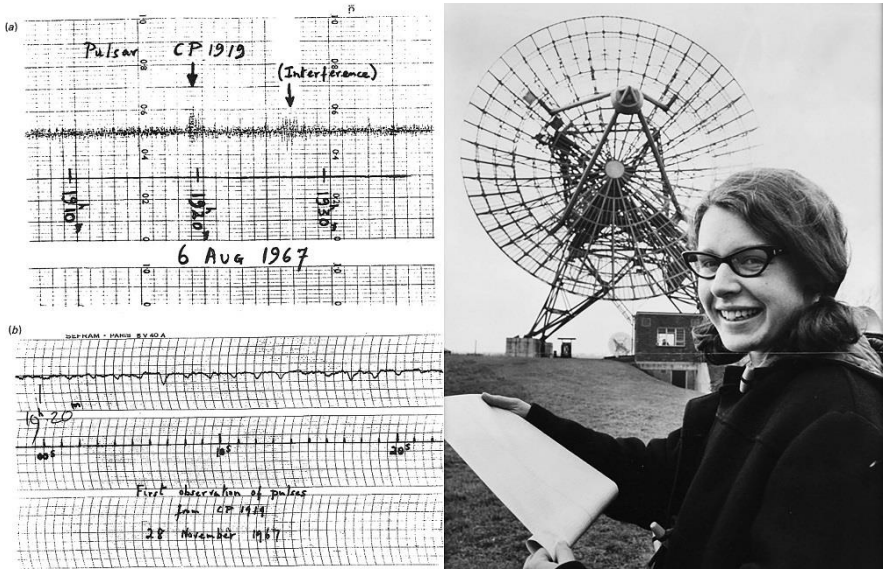
Pulsars

- Rapidly rotating neutron stars that generate radio emission.
- First identified in the 1960s by Jocelyn Bell Burnell.
- Their rotational periods are so precise that they can be used as clocks.









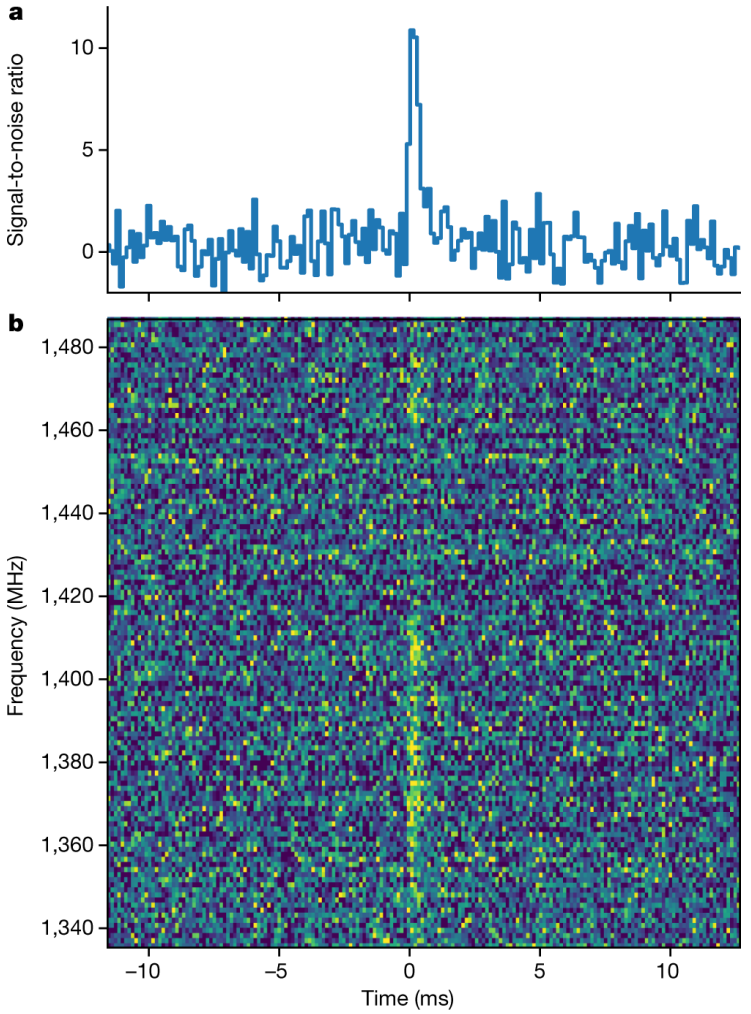
Top: The first pulsar identified

Bottom: Jocelyn Bell Burnell with the data track and the Jodrell Bank Telescope in the background

Taken from the Jodrell Bank Telescope website

Fast Radio Bursts (FRBs)

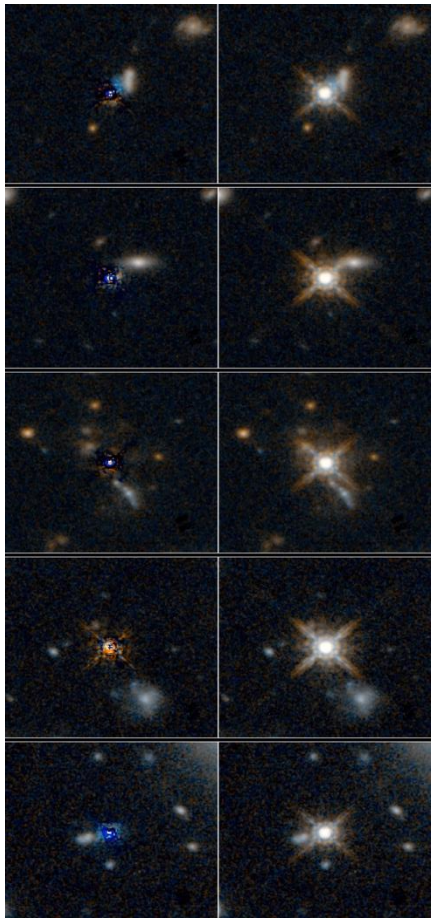
- Incredible bursts of energy where an object becomes a really bright source of radio emissions in the sky for a few milliseconds.
- Important information regarding the material that exist along the line-of-sight.



Credit: Ravi et al. (2019) *Nature*

Quasars, Radio Jets, SMGs and AGN

- Quasars are bright objects first discovered in radio surveys which looked like point sources.
- We now know that these are galaxies with Active Galactic Nuclei (AGN) lurking in their cores.
- Sub-milimeter Galaxies (SMGs) are galaxies with bright radio emission.
- Understanding these objects is fundamental to studying the [illegible]iverse.



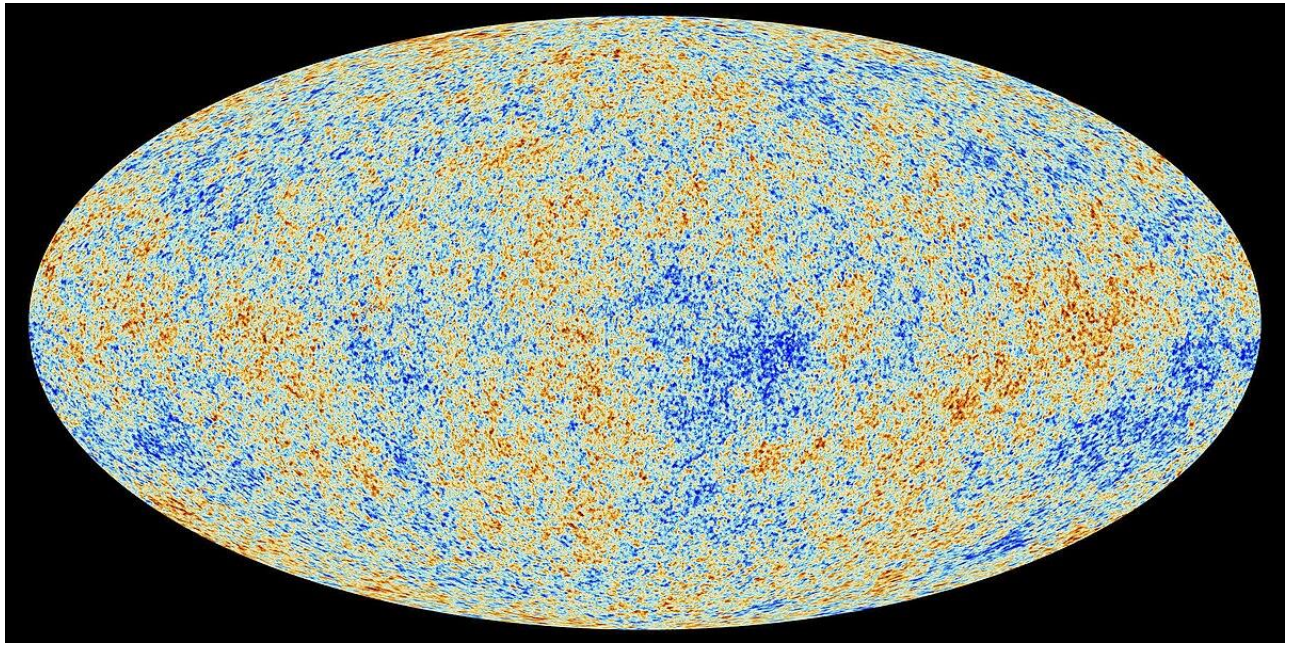
Some quasars seen with the *Hubble Space Telescope*. Credit: NASA/ESA



Centaurus A in radio with the MWA. You can see jets emanating from the galactic core. Credit: McKinley et al 2022 *Nature*

Cosmic Microwave Background (CMB)

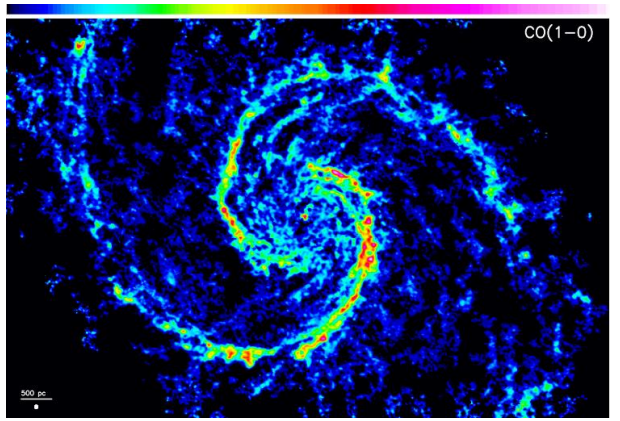
- The remnants of the Big Bang, it was discovered when scientists at Bells Lab thought the static from their antenna was caused by pigeon droppings.
- The CMB is an important confirmation of the Big Bang Theory, and inhomogeneities in it led to the introduction of the idea of inflation.



The highest resolution image of the CMB. Credit: Planck Collaboration and ESA

Molecules in space

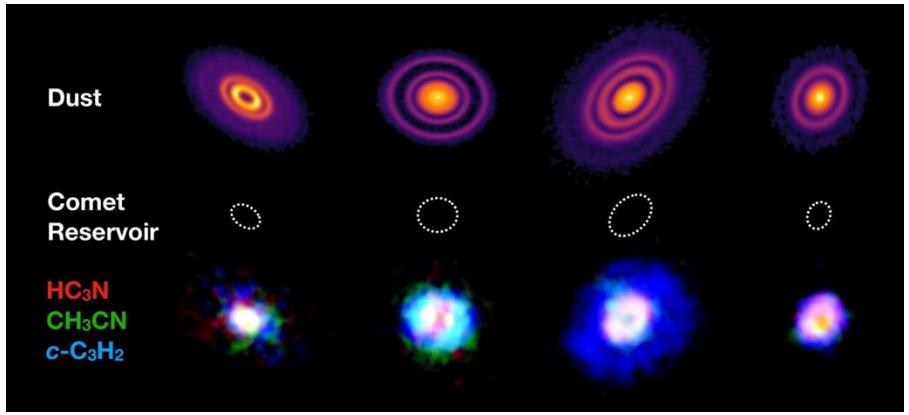
- Spectroscopic observations using radio telescopes have been essential
- Cool (pun intended) gas clouds, the ingredients of stars, have atoms/molecules with spectral lines in the radio
- Recent facilities like ALMA have tremendously increased the number of molecules found in space, both nearby and in early universe



VLA CO(J=1-0) image of M51
Credit: Schinnerer et al. 2013



Optical *Hubble* image of M51



Four young solar systems imaged by ALMA. Comet reservoir at 50 AU. Credit: J.D. Ilee/Univ of Leeds

Radio Telescopes: the hardware

Curtin University

Antennas

- Radio Telescopes, unlike their optical counterparts, use antennas.

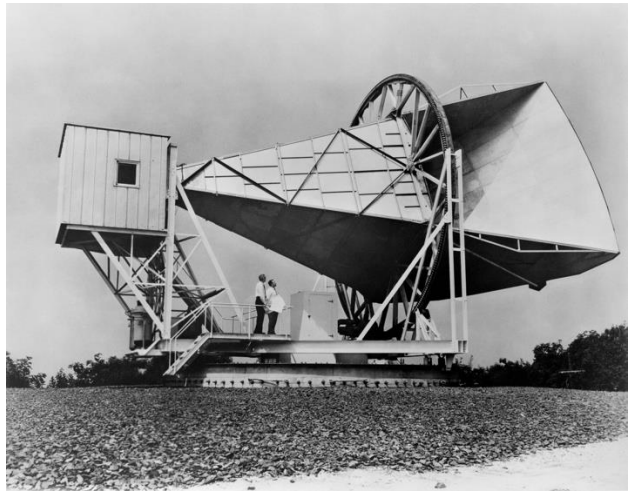
"Murriyang", with a parabolic dish



Antennas of the Long Wavelength Array



Holmodel Horn Antenna



The IRAM 30m telescope, a filled parabolic dish



The problem of resolution

- The ”resolving power” of a telescope increases with larger diameter of scope, but decreases with longer wavelengths ($\theta_{min} \sim \frac{\lambda}{D}$).
- A radio telescope observing at a wavelength of 21cm would need to be approximately 2km(!) in diameter to roughly match the resolving power of our eye observing in the visible range.
- Mechanical constraints make [illegible]opes difficult to build.





This 500m aperture telescope has a resolution of only ~9 arcminutes at 21cm; *Hubble* has ~0.1 arcsec resolution at 500nm!

Solution: Interferometry

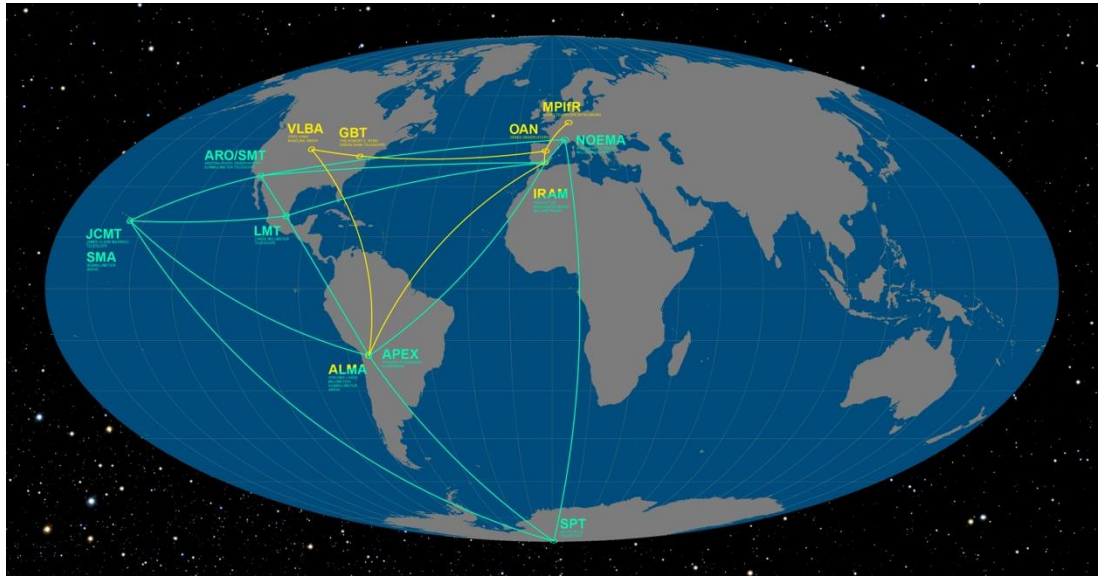
- Instead of building large telescopes, you can build multiple smaller antennas and spread them out.
- Each pair of antennas (called a “baseline”) samples independent data points.
- You can reconstruct an image with greater resolution than possible with just one antenna.



The VLA

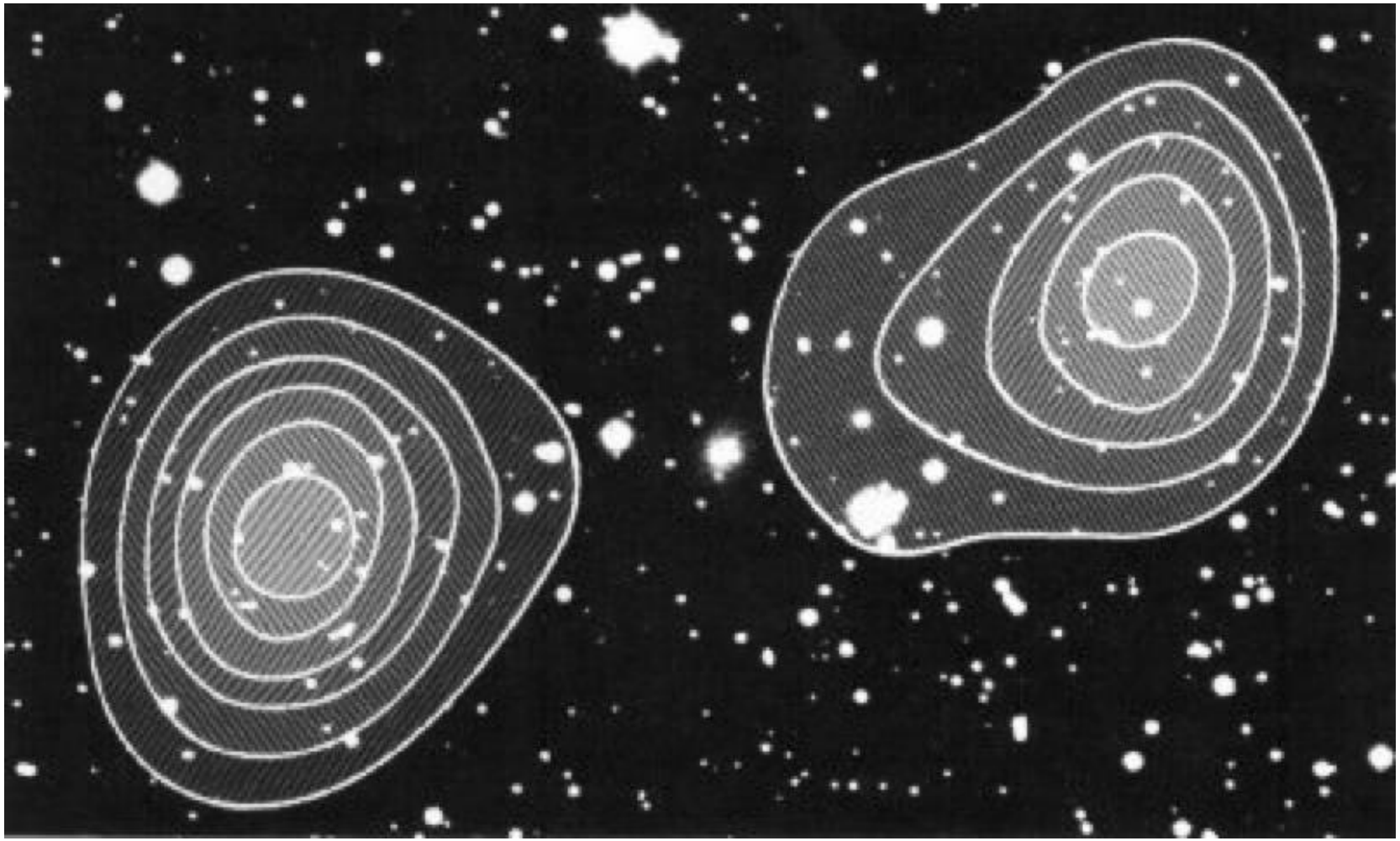


ALMA



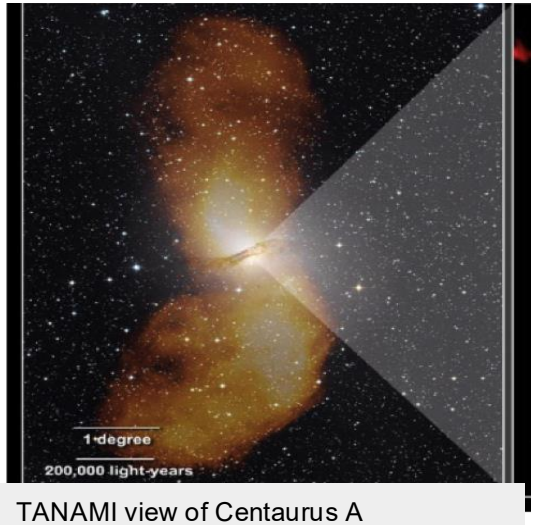
EHT

Radio Image of Cygnus A at the time of
Hewish, Ryle and Reber circa 1960s

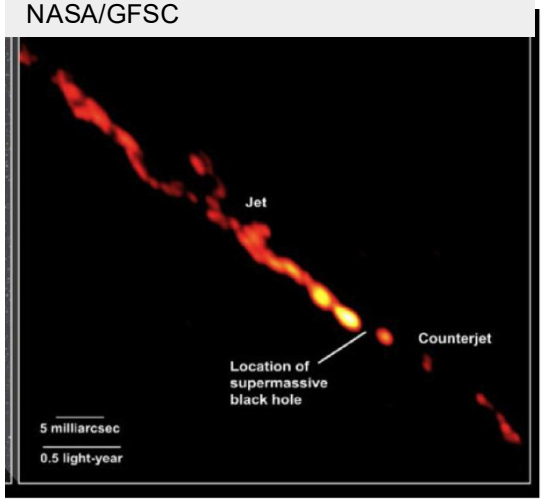


VLBI (Very Long Baseline Interferometry)

- You can create telescopes the size of the Earth by using very long baselines;
- Requires very precise timing arrays or physically transporting hard drives.
- The physical scales studied very small; it would be like being able to see a tennis ball on the surface of the moon.



TANAMI view of Centaurus A
© C. Müller (Uni-Würzburg) & NASA/GFSC





Radio Astronomy research today

- Theoretical, experimental and observational, with some intersectionality
- Lots of observational data; an hour of observing on the MWA creates roughly 60 Terabytes.
- Very collaborative
- Very interdisciplinary


First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole

The Event Horizon Telescope Collaboration

(See the end matter for the full list of authors.)

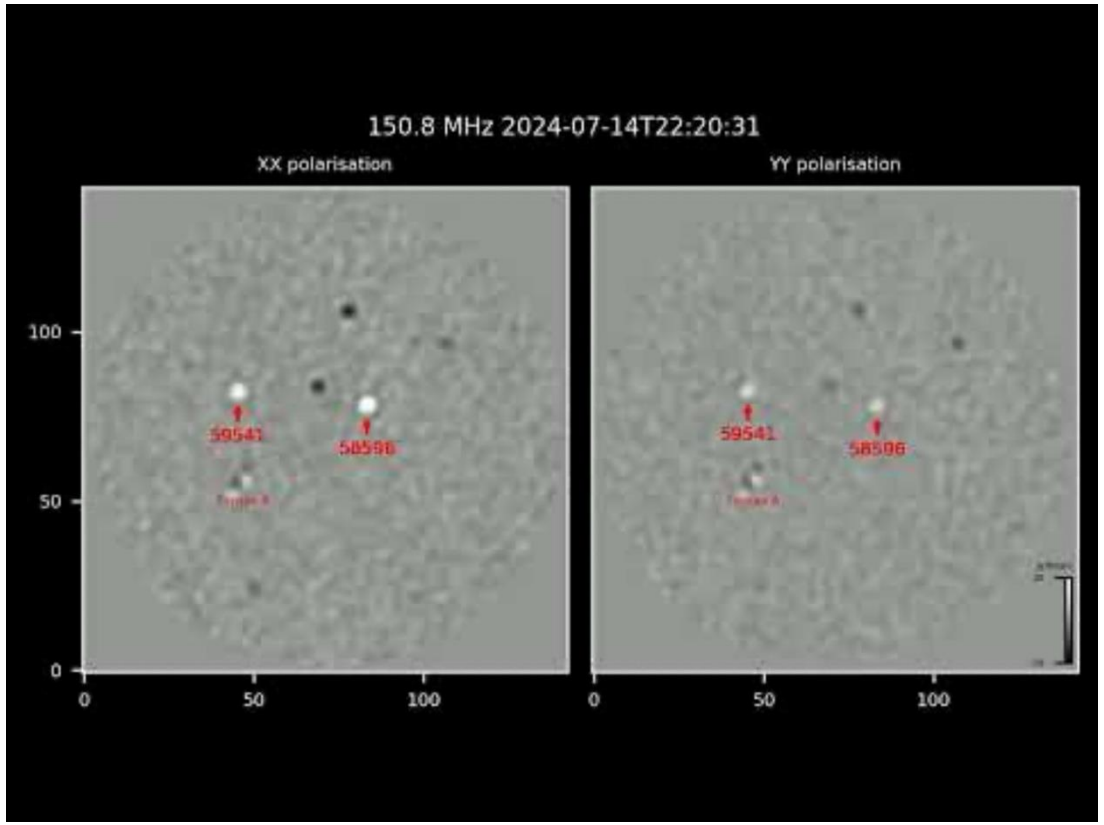
Received 2019 March 1; revised 2019 March 12; accepted 2019 March 12; published 2019 April 10

Abstract

An extreme example: list of collaborators for the first EHT paper

Modern Challenges

- Radio Frequency Interference (RFI) on the ground
- RFI from satellites
- Data storage constraints; bigger telescopes being built and conceived will have even larger data throughput



Satellites in a radio image. For context, Fornax A is one of the brightest radio sources in the sky. Video Credit: Grigg et al. 2025

What skills do you learn/acquire

- Technical Skills:
  - Physics
  - Statistics
  - Programming: command line and python are extensively used
- “Soft” Skills:
  - Presentation skills
  - Ability to do groupwork and a collaborative spirit
  - Grit—the ability to persevere through setbacks and challenges
  - Intuitive estimation: the ability to intuit/work out simple orders of magnitude



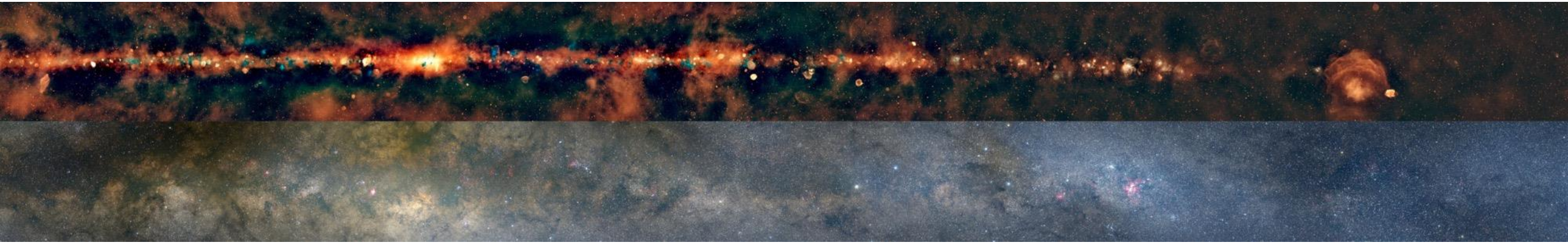
ICRAR and the Curtin Institute of Radio Astronomy (CIRA)

- ICRAR was established in 2009 to help build the science case for the Square Kilometre Array (SKA), whose low-frequency component is being built in Western Australia.
- Two nodes, one at the University of Western Australia (UWA) and at Curtin University, both based in Perth.
- Focus areas: science, engineering, data-intensive astronomy and translation and impact.
- Presence in Singapore at the Curtin Singapore campus

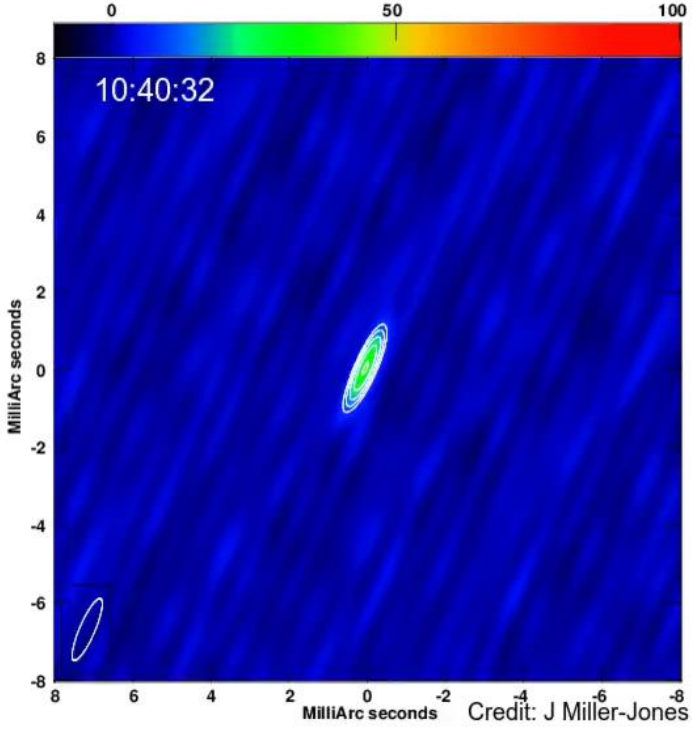


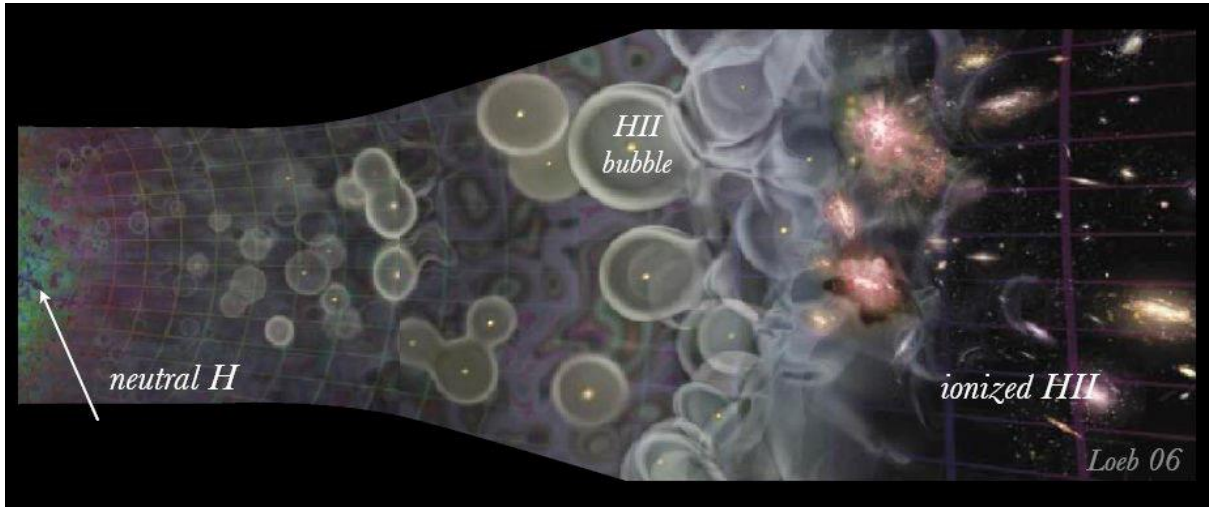


Astrophysics at CIRA



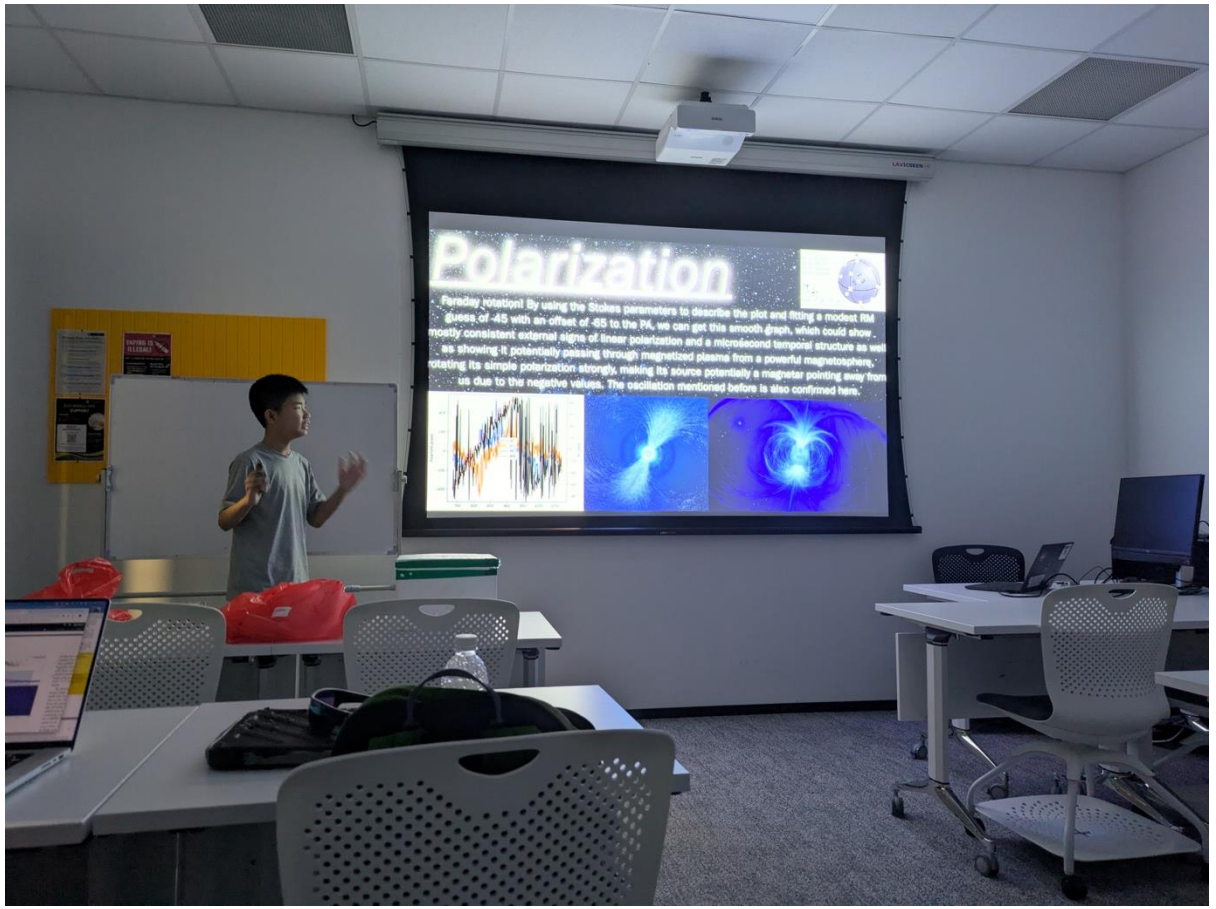






Curtin University

Recent Workshop on Radio Astronomy



Curtin University

Acronyms

- ICRAR: International Centre for Radio Astronomy Research
- IRAM: Institut de Radioastronomie Millimetrique
- SKA: Square Kilometre Array
- NRAO: National Radio Astronomy Observatory
- NARIT: National Astronomy Research Institute Thailand
- CASA: Common Astronomy Software Applications
- CSIRO: Commonwealth Scientific and Industrial Research Organization
- FAST: Five-hundred-meter Aperture Spherical Telescope
- ATCA: Australian Telescope Compact Array
- ASKAP: Australian Square Kilometre Array Pathfinder